

The Effects of the 2018 Trade War on the Economy: A Review of the Literature

Benjamin Abshire[†]

May 1, 2025

Abstract

I review the current economic research on the effect of the 2018 US-China Trade War on the United States' economy. Specifically, impacts on welfare, consumption, employment, trade reallocation, investment, imports, and exports are considered. The literature found vast, negative impacts on the U.S. economy. The welfare loss from the trade war was around 0.08 percent of real GDP in 2018 and 0.13 percent of real GDP in 2019. Import tariffs failed to return jobs to the U.S. at a meaningful scale, and aggregate investment decreased by 1 to 2 percent. Both imports and exports fell, and there is substantial evidence of trade reallocation. Contrary to traditional trade theory, there was a near-complete pass-through to U.S. import prices, implying that the incidence of the trade war fell on the United States. Finally, near complete pass-through extended to consumer goods, though not consistently.

[†]ECON 42191 - Senior Seminar in Economics. Instructed by Dr. Kathryn Wilson. Kent State University.
Email address: babshire@kent.edu

I Background

Starting in 2018, the Trump Administration imposed tariffs on a wide range of products and countries, citing three distinct reasons why. There were concerns over washing machines and solar panels, steel and aluminum, and technology and intellectual property (Bown and Kolb 2025). In response, several governments—including China, Canada, Mexico, and the European Union—responded with their own retaliatory tariffs, triggering a trade war that was, at that time, unprecedented in the modern global economy. The bulk of the trade war was fought with China, though several tariffs were put in place on traditional allies of the United States. At its height, 17 percent of U.S. imports were covered by protections totaling over \$420 billion (Fajgelbaum 2020b). When the Biden Administration began in 2021, many of the tariffs were lowered or rescinded; however, some tariffs, especially those against China, have remained in effect (Bown and Kolb 2025). What were the economic consequences of this trade war on the United States? In the following sections, I review the literature on the tariffs’ effect on welfare, employment, trade reallocation, investment, imports, and exports.

II Welfare

One of the first papers to examine the welfare effects of the trade war was Amiti et al. (2019). It estimated the effect of the 2018 tariffs on pass-through, dead-weight loss, prices, U.S. exporters, and imported varieties. The authors used visual methods and regressions throughout the paper. To estimate pass-through and domestic import prices, customs data was gathered at the ten-digit level of the Harmonized Tariff Schedule (HTS10). Import values were divided by quantity to get the unit values for each good from a country; which estimates foreign export prices since the unit values are computed before tariff payments. Multiplying by the tariff rate, the authors found the tariff inclusive price paid by U.S. importers. Then, goods affected and unaffected by tariffs were plotted with respect to price and time. In a formal analysis, assuming tariffs are exogenous, Amiti et al. regressed the log change

in foreign exporter prices on the change in one plus the tariff on imports over a year-long period. The same was done for the log change in import quantities and the log change in import values. To calculate dead-weight loss, Amiti et al. assumed the demand curve for imports has a constant slope. Then, they used data from previous regressions to define a right triangle with height of the size of the tariff and width of the change in import quantities. Finally, calculating the area of the triangle gave an estimate of dead-weight loss. Similar regressions were used to estimate impacts on U.S. exporters and producer prices, along with the visual approach described above to estimate effects of tariffs on import varieties.

Visual analysis by Amiti et al. found no change in goods unaffected by tariffs but increases similar to the size of the tariff rates in goods that are affected. In regression analysis, a variable measuring the log change in foreign exporter prices had a coefficient of -0.012. This led Amiti et al. to conclude that foreign exporters did not lower their prices in response to tariffs, causing an almost complete pass-through of the tariffs onto U.S. importers and consumers. Further, a 1 percent increase in tariff rates caused a 6 percent decline in import quantity, while import varieties of goods targeted by tariffs also fell. Amiti et al. found a cumulative dead-weight loss of \$8.2 billion in 2018; the authors noted that since there was near complete pass-through, the \$15.6 billion in tariff revenue came directly from U.S. citizens in the form of increased taxes. Depending on whether the government used this money to improve welfare by an equivalent amount, losses may be larger. The total aggregate loss was equal to about 0.08 percent of GDP. Producer prices also increased; for tariffs applied to an industry's inputs, it was found that for a firm that imports 15 percent of its inputs, a 10 percent tariff results in a 2.9 percent increase in its prices. There was incomplete pass-through because firms absorb some of this cost in their profit margins. Further, tariffs meant to protect an industry's output result in domestic firms raising their prices. Turning to U.S. exporters, it was found that a 1 percent increase in foreign tariff rates resulted in a 3.9 percent decline in U.S. export values. Finally, U.S. exporters were

found to have not lowered their prices, implying pass-through of foreign retaliatory tariffs to their own consumers.

Further estimates of the effect on welfare were provided by [Fajgelbaum et al. \(2020a\)](#). They considered dead-weight loss, tariff pass-through, import quantity, and regional effects of tariffs on real wages at the county level. All effects are estimated for tariffs implemented in 2018. Then, in an update using data from 2019, [Fajgelbaum et al. \(2020b\)](#) looked at the same aggregate effects to see if the effects of the trade war changed over a longer time frame.

Fajgelbaum et al. gathered trade data from several sources, both foreign and domestic, at Harmonized System (HS) levels 6, 8, and 10, along with county level data for regional effects and defined sectors at the North American Industry Classification System industry group level 4 (NAICS-4). An event study framework was used to initially estimate pass-through. Then, a demand side model of the U.S. economy was developed to guide regression estimations of pass-through of U.S. and foreign tariffs to import prices. Next, panel data was used to develop elasticities of substitution at the product and sector levels. Last, a general equilibrium model concerned with the supply side of the U.S. economy was developed to estimate the aggregate and regional effects of the trade war. The model assumed perfect competition, flexible prices, and adjustment of intermediate inputs. Effects on aggregate real income were estimated using the aggregate equivalent variation function EV . The function sums the increase in tariff-inclusive costs of the pre trade war import basket, the increase in the value of the pre trade war export basket, and the change in tariff revenue. When estimating regional effects, the model was changed to match specialization patterns across U.S. counties.

Fajgelbaum et al. first looked at import values and quantity through the event study framework. Import values decreased by 20 percent while quantity decreased by a similar 23 percent. Ex-tariff prices do not change, however, duty inclusive prices rose for goods affected by tariffs; which implied complete pass-through to U.S. import prices. Similar re-

sults were found for U.S. exports, implying complete pass-through to foreign import prices. Estimates guided by the demand side model confirm that of the event framework, strengthening the finding of pass-through to U.S. import prices. Additional regressions found that U.S. exporters did not lower their prices, however, they experienced a 9.9 percent reduction in export values. Calculations of product elasticities revealed that the decline in imports due to tariffs was not offset by imported goods from countries not affected by tariffs. The above results hold after several robustness checks, including heterogeneity of pass-through rates and time horizon concerns. Initial calculations assuming complete pass-through found a consumer surplus reduction of \$50.8 billion. The *EV* function found a dead-weight loss of \$7.2 billion. Regarding regional effects, county level exposure to U.S. tariffs and foreign retaliatory tariffs was estimated. The industrial regions of the Midwest and Northeast received the most protection from U.S. tariffs, while agricultural areas were targeted more by foreign retaliatory tariffs. Reduction in real wages was highest in the latter areas. Also, U.S. tariffs protected politically contested counties, while foreign retaliatory tariffs targeted counties that were heavily Republican. Workers in Republican counties had the largest costs of the trade war, seeing the biggest reduction in real wages. The authors concluded with a note that their estimations do not consider pass-through to U.S. retail prices, uncertainty impacts on business, country level wage effects, and long run impacts of the trade war. In an update using 2019 data, [Fajgelbaum et al. \(2020b\)](#) found evidence of continued complete pass-through. Further, the decline in both imports and exports persisted, along with a dead-weight loss of \$24.8 billion from 2018-19.

The question of retail price pass-through was answered by [Cavallo et al. \(2021\)](#). The authors also expanded on the findings of near complete pass-through to import prices, the response of U.S. exporters to foreign retaliatory tariffs, exchange rate pass-through (ERPT), and front-running and trade diversion techniques used by 2 major U.S. retailers. A similar approach to [Amiti et al. \(2019\)](#) was used to gather import data to determine pass-through:

Bureau of Labor Statistics import data was collected to determine U.S. import prices, and with HS codes, Cavallo et al. was able to see what tariff rate was applied to each good over time. Then, a visual approach plotted the tariff inclusive log price indices for 7 constructed groups of U.S. imports over time. These groups included imports not affected by tariffs, imports from China affected by tariffs, and imports affected but not subject to tariffs as they are imported from elsewhere. In a regression analysis, the log change in ex tariff prices of goods was regressed on sectoral inflation rate controls along with point estimates that give the estimated tariff pass-through rate and ERPT one year after implementation. This regression allowed for estimations of the effects tariffs had on U.S. import, export, and retail prices. To better understand the effect on retail prices, price data on several goods affected by tariffs was gathered and their price indices were plotted with time. For a fuller picture, data from two major U.S. retailers totaling over 90,000 goods was gathered and plotted in similar patterns as before. Then, a regression similar to the one described above was used to estimate the impact on U.S. retail prices. To study trade diversion and front-running, tonnage imported from China and elsewhere by two firms was plotted with time. Then, the share of this tonnage imported from China was plotted in the same way.

Cavallo et al. found near complete pass-through to U.S. import prices. Visual analysis showed no change in goods unaffected by tariffs, but a jump in price of goods affected by tariffs similar in magnitude to the tariff rates. Regression analysis revealed that a 10 percent tariff results in a 0.6 percent ex-tariff price and a 9.4 percent increase in prices paid by U.S. importers. Subsequently, ERPT estimates revealed a rise in import prices by 2.18 percent when the U.S. dollar depreciates by 10 percent, meaning the depreciation of the Chinese Renminbi did not offset the cost of the tariffs. On average, U.S. exporters did lower their prices. Regression analysis found that a 10 percent tariff reduces prices by 3.3 percent; implying a 33 percent pass-through to U.S. export prices. This suggested that some of cost of foreign retaliatory tariffs fell on U.S. exporters. Cavallo et al. attributed

this drop in price due to the fact that unlike imports, half of U.S. exports affected by tariffs are undifferentiated goods, meaning that Chinese importers can get the same good from elsewhere. Shifting focus to retail prices, visual analysis of a sample of goods affected by tariffs—washing machines, handbags, tires, and bicycles—suggested that pass-through was not consistent across all goods. Some goods like washing machines saw immediate and complete pass-through to prices, while other goods like tires did not see effects until tariffs were increased to 25 percent. Looking at data from the two major U.S. retailers, a 10 percent tariff on Chinese goods resulted in a 0.35 percent increase in price relative to unaffected goods. What was the reason for the heterogeneity in pass-through between import and retail prices? Cavallo et al. theorized that U.S. firms absorb the costs in their profit margins, implying that U.S. firms bore the cost of these tariffs. U.S. firms increased imports before tariff implementation (front-running) and imported some goods from elsewhere after implementation (trade diversion). This led Cavallo et al. to conclude that they observed only the short run response to tariffs. As time goes on, Cavallo et al. predicted pressure on U.S. retailers, resulting in a combination of lower ex-tariff U.S. import prices or increased pass-through to retail prices.

What happens when there is pass-through to retail prices for a particular good? A case study by [Flaaen et al. \(2020\)](#) looked at the effects of the 2018 tariffs and the 2012 and 2016 anti-dumping duties on washing machines and dryers. The authors chose washing machines due to the range of trade protections placed on them, the availability of data—they were targeted by one of the first tariffs implemented in the trade war—and the clear classifications of the good. A visual analysis was used to estimate trade diversion and front-running for all protections along with preliminary estimates of pass-through to import prices. To identify the price trends of washing machines, Flaaen et al. started by gathering data on the prices of 5 household appliances from different manufacturers. Then, a hedonic regression analysis was employed, where the log price of an appliance was regressed on fixed effects that account

for retailers, product mix, product features, and product life cycle. A similar approach was used to estimate the effects of the anti-dumping and tariff policies. To isolate the effects of the tariffs, the price trends of washing machines versus other appliances were factored in. Since tariffs were not applied to U.S. manufacturers, the magnitude of the price increases by foreign manufacturers could be masked. To explore the effects by brand, a similar regression to those above was employed that shows separate coefficients for each brand. Finally, the Tariff Elasticity of Consumer Prices (TECP) was used to measure the price changes in goods relative to the change in tariffs. It shows the change in expenditure by consumers that is required to obtain the same utility as before the tariffs were implemented. To calculate the total cost to consumers, the increase in prices was multiplied by the number of shipments in a given year.

Visual analysis of imported washing machines by country showed that once anti-dumping duties were levied, imports from unaffected countries increased substantially, which suggested that U.S. importers engaged in trade diversion. Once the global tariffs on washing machines were announced, there was a large increase in imports before it went into effect, suggesting that U.S. importers engaged in front-running. Additional visual analysis showed that during the 2012 anti-dumping duties, the price of washing machines fell. However, once the tariffs were levied, prices rose. Notably, the prices of dryers, which were not affected by tariffs, also rose in the same way. Hedonic regressions found that the 2016 anti-dumping duties caused a modest increase in price of the two appliances of about 3.5 percent in a 8 month period. The sample was not long enough to estimate an effect from the 2012 duties. Once the tariffs were implemented, a price increase in both appliances of 11 percent was found. A similar increase of 10 percent was found with wholesale retailers. Why did the prices of dryers increase? Flaaen et al. found that it was due to washers and dryers being close complementary goods. In fact, they are often sold together, so it was possible that firms spread the cost of the tariffs across the two goods. Looking at the price effect of different

manufacturers, they found that all manufacturers increased prices by about 5 to 17 percent. Notably, this included domestic manufacturers who were found to be using their market power to raise prices. The TECP for the anti-dumping duties was -1.62 for 2012, implying a negative pass-through and therefore a decline in price, and 0.31 for 2016, implying a small increase in price, and incomplete pass-through. For the 2018 tariffs, the TECP ranged from 1.25 to 2.26 using the average tariff rate. This implied complete pass-through to consumer prices and a further increase in price by domestic manufacturers. Finally, the total cost to consumers was found to be \$1.55 billion annually. The tariffs brought 1,800 jobs to the US, but at an expense to the consumer of about \$817,000 per job.

A final update on welfare effects using data from most of 2019 was done by [Amiti et al. \(2020b\)](#). They looked at pass-through rates, terms of trade, and the redirection of trade. The same approach to gathering data was used as in [Amiti et al. \(2019\)](#). Then, the data was separated into categories of capital goods, consumer goods, and intermediate inputs. Intermediate inputs were further divided into steel and non-steel inputs. The authors used a difference in differences estimation where they regressed log import prices on the interaction between a month indicator and the log change in tariffs, country-time fixed effects, HTS10 time fixed effects, and country-product fixed effects. The first difference was between countries with and without tariffs, and the second difference was before and after the tariffs were levied. Since the regression included log measures of different variables, the coefficients are elasticities.

Estimated elasticities for the U.S. import prices of the categories described above jumped to 1 and stayed there after tariffs went into effect, implying a complete pass-through to U.S. firms and consumers. However, Amiti et al. noted that this was not the case for steel inputs. After a complete pass-through, the rate fell to around 50 percent after 1 year. The authors found that overall, Chinese exporters had still not lowered their prices through October 2019, where the data ends. Further, U.S. import volumes were found to drop after tariff

implementation and fall faster as time goes on. This led Amiti et al. to find that firms take time to reorganize their supply chains to avoid tariffs. Steel inputs were again an exception here, the authors found that since foreign firms absorbed the tariff cost, steel inputs fell less relative to other sectors.

Did the trade war have an effect on consumption? [Waugh \(2019\)](#) looked at this question directly. He measured the elasticity of consumption growth from Chinese retaliatory tariffs and its effect at the county level. Specifically, automobile sales and retail employment were used as proxies to measure changes in consumption. Waugh also measured the change in exports to explain the effect on consumption and provided estimates of impacts on employment.

Waugh started by describing models of real wages and consumption per capita within the labor market. Potential shocks to consumption result from changes in earnings and expectations about the future along with the distribution of asset holdings. Data on goods affected by Chinese retaliatory tariffs was gathered at the HS10 level and was used to build a county-level measure of the exposure to Chinese retaliatory tariffs. The measure reflected high tariff exposure if a county had high rates of employment in sectors affected by retaliatory tariffs. To measure consumption, Waugh used new automobile sales and retail employment as proxies. Retail employment was included as a measure of non-durable goods, since expectations about the future can greatly effect the consumption of durable goods such as automobiles. Two other measures of employment were considered: goods producing and total employment. To measure exports, trade data from the U.S. Census was gathered and used to construct a measure of a county's exports on a per worker basis. Visual analysis and a difference and differences estimation was used to determine effects on automobile sales. The 12 month log difference in automobile sales was regressed on the 12 month difference in tariff rates and various fixed effects. A similar regression was used to ascertain effects on county level exports and retail employment. To address concerns over a possible violation of

the parallel trends assumption, a regression was ran that checked for any preexisting trends, announcement effects, and shocks that may be more associated with rural or manufacturing counties.

Visual analysis revealed that before the tariff implementation, consumption as measured by automobile sales grew the by same amount in low and high tariff counties. After tariffs were levied, high tariff counties grew 2 percent slower than low tariff counties. Regression analysis found that moving from the least affected to most affected counties resulted in an approximately 3.8 percent decrease in consumption. A similar analysis on county level exports found elasticities ranging from -10 to -21. This implied that exporters cannot export their products elsewhere, leading to welfare loss. Considering employment, Waugh found that moving from the least to the most affected counties yields a 0.70 percent decline in retail employment along with a 0.75 percent decline in total employment and a 1.70 percent decline in goods producing employment. Robustness checks for the parallel trends assumption confirmed earlier results of a decline in consumption as measured by automobile sales. Further, sales started their decline in March, around the announcement of the tariffs but before their implementation. This suggests that the announcement of tariffs caused some to stop purchasing cars out of uncertainty. Similar results were seen in the three measures of employment, though the anticipatory effects were not as clear as in automobile sales. The combination of the above results supports the parallel trends assumption. Back of the envelope calculations revealed that retaliatory tariffs resulted in about 67,000 lost automobile sales, a 0.39 percent decline in total sales. Assuming the average purchase price of a car, the lost automobile sales totaled to \$2.5 billion by January 2019. Using the automobile estimates with the Consumer Expenditure Survey, Waugh estimates a reduction in aggregate consumption of 0.28 percent and 0.39 percent (\$40 billion and \$54 billion). Turning to retail employment, Waugh found a reduction of about 22,500 jobs, a 0.14 percent decline (\$20 billion reduction in aggregate consumption). Waugh noted that the aggregate effects

found above are concentrated on the high tariff counties, implying high concentrated welfare losses. These analyses found that Chinese retaliatory tariffs reduced a county's exports and employment, which then reduced consumption. Waugh also stated that his analysis does not include the intensive margin, like reductions in hours or wage cuts. A more formal economic model was thought to be needed in order to account for the durable nature of the consumption data, the joint relationship between exports, employment, and consumption, and expectations of the future.

III Employment

In the previous section, [Waugh \(2019\)](#) used measures of employment as a proxy measure for consumption and found that Chinese retaliatory tariffs caused a small reduction in total employment. What were the effects on the U.S. manufacturing sector? [Flaaen and Pierce \(2019\)](#) consider the effects of U.S. tariffs on steel, inputs, and foreign retaliatory tariffs. Specifically, Flaaen and Pierce look at the tariffs' effect on employment, output, and producer prices.

Data for the analysis was gathered from several sources, including the United States Trade Representative, the U.S. Federal Register, Canadian Department of Finance, European Commission, World Trade Organization, U.S. International Trade Commission, Annual Survey of Manufactures, The Federal Reserve, and the Bureau of Labor Statistics. HS codes were used to map tariffs to the NAICS 4 level. Flaaen and Pierce then identified 3 measures that capture the changes in tariffs on factors that could drive employment effects on the manufacturing sector: import protection in the U.S., foreign demand for U.S. exports, and U.S. input costs. The measure for import protection was constructed by dividing the sum of an industry's imports subject to tariffs by the industry's output plus the industry's imports minus exports. Next, the authors constructed a measure for the foreign demand for U.S. exports, since foreign retaliatory tariffs affect U.S. manufactures' competitiveness overseas.

It divides the sum of U.S. exports subject to retaliatory tariffs by an industry's output. A measure of input costs was constructed by multiplying the sum of the share of input costs from a foreign country by the import share of the domestic supply of a commodity, in a given industry. The measure aimed to capture the effect of U.S. tariffs on domestic input costs by using their supply chain linkages with foreign countries. Then, a series of difference in differences estimations were used to determine individual effects of tariffs on the manufacturing sector. Log employment, log output, or log of the Producer Price Index was regressed on the industry measures, time controls, and industry controls. To address the potential issue of differing trends pre-tariff, Flaaen and Pierce differed out pre-trends for each coefficient.

Results from the import protection measure revealed that the tariffs on steel, aluminum, and washing machines resulted in the most protection for those industries. Foreign retaliatory tariffs resulted in increases of tariffs on less than 10 percent of domestic manufactures exports. Input costs rose the most for industries that are dependent on metals for production. Regression analysis revealed that rising input costs and foreign retaliatory tariffs had a negative impact on employment. While import protection had a positive effect, it was outweighed by the other two measures. In total, moving from the 25th to 75th percentile of exposure to retaliatory tariffs causes a reduction in employment of 1.4 percent. There was little change in manufacturing output, suggesting that production was not affected by any of the tariffs. Finally, rising input costs caused producer prices to rise, while effects from import protection and foreign retaliation were insignificant. It was noted that these are the short-run effects of the tariffs and longer run effects may be different. It was theorized that if the tariffs remained in place for an extended period of time, more manufacturing could be brought to the U.S., but at a substantial cost to consumers. Flaaen and Pierce concluded that positive effects from tariffs designed to protect domestic imports were offset in the short run by higher costs and reduced competitiveness from foreign retaliatory tariffs.

Did subsidies given to farmers offset job losses? [Autor et al. \(2024\)](#) answered this

question along with estimating the broader effects of import and retaliatory tariffs on employment. Furthermore, the authors looked at the political consequences of the trade war on the 2020 election. Data was measured at the Commuting Zone (CZ) level, which allowed for an estimate of the effects on local labor markets. Data on employment, trade volumes, manufacturing production, tariff exposure, agricultural subsidies, and political outcomes was gathered from numerous different sources. Major sources of data include Quarterly Census of Employment and Wages (QCEW), U.S. Census Bureau’s Foreign Trade Statistics, [Fajgelbaum et al. \(2020a\)](#), EWG Farm Subsidy Database, and Congressional Election Study (CES). Autor et al. then used a 3 step process to estimate a CZ’s tariff exposure. Values were estimated for 2 equations that measure an U.S. industry’s exposure to import and retaliatory tariffs. Then, these values were input into an equation that estimated the effect of the tariffs on demand for U.S. outputs. This allowed for an equation measuring changes in demand for outputs of U.S. commuting zones to estimate exposure. This measure also captured spillover effects to industries not targeted by tariffs. A preliminary analysis on the effect on employment was used to see if there was any evidence to warrant subsequent analysis. The authors employed an event-study that measures the difference in seasonally adjusted employment ratios relative to a baseline value. The measure showed how long it took for employment effects to reach their maximum effect. A formal regression analysis was employed to more accurately estimate the impacts on employment. Changes in employment population rates was regressed on a CZ’s exposure to import and retaliatory tariffs, agricultural subsidies, and controls for linear trends and time. To discern effects on the sector level, QCEW data was used to divide employment into 8 different sectors and was then regressed on the same variables as in the original regression. Next, regressions using data from the CES were employed to learn of any political effects. First, the authors looked at whether support for the tariffs varies with geographic exposure to the tariff policy. Second, they looked at effects of tariff policy on party identification. Third, the Republican vote share in

a CZ was analyzed to measure electoral outcomes.

Preliminary analysis revealed that there was a small decline in employment in CZ's more exposed to import tariffs while those more exposed to foreign retaliatory tariffs saw much larger declines. The authors noted that employment rates were declining before tariff implementation, so it is difficult to say that there is a casual relationship between the two. Regression analysis revealed a weak positive effect on employment in areas exposed to import tariffs, a robust negative effect in areas exposed to foreign retaliatory tariffs, and a weak positive effect from subsidies. Multiplying the estimates by the average CZ's exposure to tariffs, this results a 0.110 percent gain in employment from import tariffs, a -0.115 percent loss from retaliatory tariffs, and a 0.028 percent gain from subsidies. This led Autor et al. to conclude that the effects from import and retaliatory tariffs are almost offsetting, and the gain from subsidies did little to offset the negative effects of foreign retaliatory tariffs. Regarding sector specific effects, import tariffs did not have an effect on manufacturing employment. It was instead concentrated in the business sector; however, this was an imprecise estimate. Retaliatory tariffs reduced employment in a number of sectors, including agriculture, transportation, and warehouse services. The subsidies did have a positive effect on employment in agriculture and business services. This led to the conclusion that at best, the trade war had no effect on employment and may have been slightly negative. The authors note that since import tariffs had no apparent effect on employment in the manufacturing industry, the intended goal of the Trump Administration to bring manufacturing jobs back to the U.S. did not come to fruition. Why was this the case? One reason may be due to trade diversion. Since U.S. imports from China have relatively low trade cost elasticities, buyers were able to import goods from places without tariffs, instead of from domestic manufacturers. Looking over the long term, while attempting to control for the effects on employment from the COVID pandemic, the authors found no change in employment due to either import or retaliatory tariffs from 2019-2021. Considering political effects, people in highly protected

CZs became more likely to vote for President Trump in 2020. Negative effects from foreign retaliatory tariffs were less effective in reducing Republican support than import tariffs were by raising support. The increased support for a party whose goal failed led the authors to conclude that overall, voters welcomed the protections despite their economic cost.

IV Trade Reallocation

Several of the papers reviewed above, including [Autor et al. \(2024\)](#), [Cavallo et al. \(2021\)](#), and [Flaaen et al. \(2020\)](#) show cases where firms engaged in trade diversion to try and get around tariffs. Is this result consistent across the board? To answer this question, I review [Dang et al. \(2023\)](#) who looks at the trade diversion effects on goods targeted and not targeted by tariffs. The authors identified three mechanisms by which firms divert trade: revealed competitive advantage (RCA), capital intensity and abundance, and colocation. Colocation occurs when exporters expand their market share of goods that are non-targeted but similar to targeted goods by entering new markets.

Data for this analysis was gathered from UN Comtrade, NBER, World Bank, and the U.S. Census Bureau; it was measured at the HS 6 and 8 levels. U.S. industries were divided into 18 separate sectors. These sectors ranged from processed food to works of art. To visualize the effect of the trade war on total import value from China by industry, Dang et al. estimated the difference of the log import value of targeted products between the first half of 2018 and 2019. Then, a visual analysis was used to show the average change in import value of targeted and non-targeted goods imported from China and the rest of the world (ROW). To obtain a better sense of diversion effects on targeted goods, a difference in differences estimation was used to measure the change in targeted goods imported relative to the number of non-targeted goods imported. To do this, the log of imports was regressed on two dummy variables and various fixed effects controlling for time, seasonal adjustment, and country-product effects. The first dummy variable becomes a 1 if a good was targeted

and imported from China and the second variable becomes 1 if a good was targeted and was imported from the ROW. Estimating just the effect on non-targeted goods yields a similar regression with extra dummy variables and interactions. This general framework was used throughout the paper for different estimations.

Visual analysis revealed that the trade war resulted in either a decline or little change in imports in all but 3 industries, while imports of non-targeted goods also rose. Comparing the change in imports between China and the ROW, it was found that imports from the ROW increased, but by a smaller amount than imports from China decreased. This provided preliminary evidence of trade diversion. Moreover, an increase in imports of non-targeted goods from the ROW indicated a spillover effect from importing targeted goods. With respect to the effects on targeted goods, imports from China fell by 26 percent compared to non-targeted goods. From the ROW, imports increased by 4 percent, strengthening evidence of trade diversion. Turning to just non-targeted goods, it was found that imports of these goods from China increased by 5.9 percent and goods from the ROW increased by 3.4 percent. The authors found that this provides some evidence that Chinese firms hedge against losses by increasing exports of non-targeted goods. Furthermore, the fact that bystander countries are also increasing exports showed early evidence of colocation. Since there were investigations before these tariffs were implemented, the authors look at the effect on imports before and after this period. After the investigation started, imports of targeted goods from China rose by a large 16 percent, and from the ROW by 8.7 percent. Imports of non-targeted goods also rose, but at smaller percentages. This gave some evidence of front-running by firms to try and stockpile goods before they have to pay tariffs on them. The authors then estimated the effects across U.S. trading partners and by industry; similar results to the visual analysis for effects across industries are found. Southeast Asian countries were most affected by the trade war, since their RCA is higher compared to other countries (making their goods substitutes). Next, the authors determined if the three drivers identified above are in fact reasons for the

effects also listed above. They found that higher the RCA, the more the U.S. will import from that country. Additionally, the spillover effect was larger when RCA is larger. Capital intensity and abundance were found to increase the spillover effect, since these countries have more available assets to change production. Regressions revealed significant colocation effects during the trade war. In fact, for the larger the RCA and size of the targeted product, there was a larger colocation effect on similar but non-targeted goods. Finally, Dang et al. found that ROW countries also exported more of both goods to other bystander countries.

Additional estimates of trade diversion were given by [Nicita \(2019\)](#). Nicita looked at effects of the U.S. tariffs on U.S. imports from China and the rest of the World, along with the response of Chinese exporters to the tariffs. Trade data was gathered from the U.S. Census Bureau at the HS 8 level and was used for both visual and regression analyses. The authors split analyses into the 2 phases of tariff implementation in the trade war. A preliminary analysis of the effects of U.S. tariffs on imports from China was conducted by comparing the change in the import values of targeted goods imported from China to those not targeted. A similar analysis was employed to view these effects across different U.S. sectors. A difference in differences estimation was used to determine the effects of U.S. tariffs on imports from China and the effects on Chinese export prices. The percent change in either the value of trade or the price of a good was regressed on dichotomous variables that indicate phase 1 or 2, a control for the relation of the dependent variable with the ROW, and controls for sectoral fixed effects. The coefficients of the dichotomous variables are the average impact of U.S. tariffs in phase 1 or 2. To measure trade diversion effects, a similar approach was used as above but instead regressed the change in U.S. imports from the ROW on interactions between the dichotomous variables and a variable that measures the correlation between imports from China and the ROW for every dollar. The coefficient on the interaction variable measures the substitution of targeted Chinese imports from elsewhere. Nicita used assumptions based on the regression results to chart data and determine the

effects of trade diversion by country across the U.S. and across sectors.

Simple visual analysis revealed that U.S. imports from China fell after the implementation of phases 1 and 2 of the tariffs. Notably, imports of goods targeted increased right before both phases were levied, suggesting the existence of front-running. Goods not targeted remained relatively stable over time. Looking across sectors, most saw a decrease in goods targeted by tariffs and an increase in non-targeted goods. The most affected sector was office machinery, with a substantial 65 percent reduction in imports from China. Regression analysis confirmed these results and found that the effect on imports from China grew with time. In phase 1 goods, the reduction grew from 22 percent to 45 percent, while for phase 2 goods, it grew from 20 percent to 25 percent. Nicita found that Chinese exporters did not lower their prices until the second quarter (Q2) of 2019 (the last time period considered in the analysis). The reduction in price of about 8 percent suggested that Chinese exporters bore some cost of the tariffs. Nicita, however, warned that based on the uncertain reliability of the data used, additional analysis with newer data is needed to corroborate this finding. The regression that estimated trade diversion found that there was a substitution of goods from other countries, but only when Chinese imports decreased due to the tariffs. In Q2 2019, goods subject to phase 1 tariffs saw trade diversion effects of 48 cents per dollar. For goods subject to phase 2, trade diversion effects were 33 cents per dollar. Further, these trade diversion effects only increased with time. In the first half of 2019, the decline in imports totaled \$35 billion worth of goods. Of this decline, \$21 billion worth of goods was replaced by other countries, and \$14 billion was lost completely. This implied that trade diversion was not enough to replace the lost imports from China. Looking across countries, Taiwan, Mexico, The EU, and Vietnam had replaced Chinese imports the most. Breaking this down by sector supported the findings of the earlier analysis; office machinery was the most affected, experiencing a reduction of \$5.5 billion worth of goods.

V Investment

Talk of changing trade policy can generate uncertainty for both firms and investors. Did the unpredictable policies during the trade war create significant uncertainty? If so, what impact did it have on capital and, therefore, investment? To find out if there was increased uncertainty, [Caldara et al. \(2019\)](#) constructed 3 measures of Trade Policy Uncertainty (TPU), one at the firm level and two at the aggregate level. Starting with the firm level construct, the authors analyzed quarterly earnings calls from publicly traded firms for words associated with talk of trade policy and uncertainty. They used instances of both types of words together to create the measure. Then, in the first measure of TPU at the aggregate level, newspaper articles beginning in 1960 were analyzed in a similar way to the firm level measure. The second aggregate measure used a stochastic volatility model for import tariff rates from 1960 to 2018. The model captured both the unexpected changes in tariff size and uncertainty about tariff rates. To estimate the effects of TPU at the firm level, Caldara et al. regressed a measure of investment using quarterly balance sheet data from Compustat on the measure of TPU, along with fixed effects for firms and time. The first regression included data from 2015 to the end of 2018, as this was the time period with the most uncertainty. It also restricted sectors to agriculture, mining, and manufacturing, as these were the only sectors that allowed the authors to construct their measure. The regression held up to various robustness checks that included only considering words associated with trade policy, dropping firm level control variables, and extending the time period back to 2005. Caldara et al. also implemented a similar difference in differences regression to that above to ascertain investment effects in 2018. They included measures of industry level capital growth changes and the change in TPU in an industry from 2017 to 2018. To estimate the effects on aggregate trade policy, the authors used 3 vector autoregression (VAR) models that included various combinations of the news based TPU, real business fixed investment, the

measure of tariff volatility, actual tariff rates, and other macroeconomic indicators. Finally, a two-country model with heterogeneous firms and a New-Keynesian open economy was used to further examine trade policy uncertainty. The model consisted of households, wholesale firms, retailers, distributors, capital goods producers, intermediate goods producers, and the government.

Firm level measures revealed a marked increase in TPU in the year 2018, with 8 of the 12 industries experiencing an increase in average TPU share, with the largest share being 30 percent in the durable goods sector. With the aggregate measures, the authors were able to reconstruct the post 1960 history of U.S. trade policy, and when comparing the news based measure to the firm level measure, the news based measure tracked well to what was observed in the firm level data. From 2015 to 2018, in response to the median value of increased firm level TPU, firms saw a 2 percent reduction in capital stock. This analysis excluded aggregate general equilibrium effects, and when separated from these effects, the authors simply aggregated the direct effect on the firm. This resulted in a finding of a \$23 billion, 1 percent decline in private non residential fixed investment. Considering effects only in 2018, the authors found that for a industry that experienced, for example, an increase in TPU that is two times the standard deviation of other industries, that industry saw a 3 percent reduction in capital growth. Analysis of the 3 VAR models revealed that, under a shock to TPU similar to that observed in 2018, aggregate investment declined 1 to 2 percent over 1 year. Finally, in the two-country model, both a higher expectation of tariffs and uncertainty about future tariffs reduced investment, GDP, consumption, and exports.

What was the effect on investment as the trade war went on? [Amiti et al. \(2020a\)](#) looked at the impact of tariff announcements on the stock market returns of firms exposed and not exposed to China to determine effects on capital. If returns to capital decline, then investment rates will also decline. Stock market return data was gathered from the Center for Research in Security Prices (CRSP) and merged with balance sheet data from

CRSP/Compustat Merged (CM) to obtain a sample of 2,864 firms across all sectors that report daily returns. Amiti et al. considered 3 different ways that firms were exposed to China: importing, exporting, and foreign sales. 46 percent of the sample was exposed to China in some way. Next, eleven instances of trade war announcements were identified and sorted into three categories: four U.S. tariff announcements, three Chinese retaliatory tariff announcements, and four other announcements. These "other" announcements allow for a placebo test to see if the US-China announcements specifically affected firms, or if it was due to other factors. Then, Amiti et al. used a modified event study of the stock market to deduce aggregate stock market effects and the impact on the returns of firms. An event's impact was broken down into two parts: the common effect (how changing a factor that matters to all firms affected returns) and the differential effect (the additional effect on firms exposed to China). This was done by nesting an event study inside a factor model, which captures firms' stock returns as related to a set of variables that may affect them. The method allowed for Amiti et al. to decompose stock returns in an event window into a component explained by the common and differential effects. The effects of the announcements were measured by using event windows that are between 1 and 30 days from when an announcement was made. To measure effects on investment, the relationship between stock prices and market to book (MTB) values was used. Lower stock prices result in lower MTB, which means lower returns to capital and therefore lower investment.

Estimations for the effects of U.S. tariff announcements in a seven day window revealed heterogeneous effects across announcement dates. When analyzing the firms who import from China, 2 of the 4 events had a negative effect on their returns, while one actually had a positive effect. Firms that had high sales in China saw lower returns than the other China exposed firms. For firms that export to China, there was no clear effect from the announcements. In total, the stock market fell 3.5 percent over the four, seven day event windows on the announcements. Moving on to announcements of Chinese retaliatory tariffs,

firms who import and export do not see many effects, with the exception of one event that had a negative impact on exporters. Firms that had high sales in China, however, saw larger declines in returns. In fact, the 100 largest firms in this category saw a 2.5 percent decline in abnormal returns. The total effect on the stock market was much smaller here, only declining by 0.8 percent over the three, seven day event windows on the announcements. Placebo effects show that the results found above were not affected by non US-China announcements. In terms of investment effects, a firm whose average MTB was 1.82 saw a drop in MTB value of 0.69 percent when stocks decreased by 1 percent. This provided evidence that the trade war did lower investment. The aggregate changes in investment did not become very apparent until Q1 of 2019. This was due to the fact that changes in stock prices do not affect investment in the author’s framework until 4 quarters later. Significant declines started in Q1 2020 and were estimated to continue all the way until Q3 of 2020, where a total decline in the investment growth rate of 1.9 percent was estimated by Q4 2020. The tariff announcements from the U.S. and China lowered stock prices of the sample by 6 percent, which resulted in an aggregate loss in firm value of \$1.7 trillion. The authors concluded by noting several conditions, including that they only estimate the unanticipated part of these announcements, so their results may have been underestimates.

VI Exports

Several of the papers discussed above found that U.S. exports fell overall. Specifically, [Waugh \(2019\)](#) found that Chinese retaliatory tariffs reduced a county’s exports. What impact did the U.S. import tariffs have? To answer this question, [Handley et al. \(2025\)](#) looked at the effects on products exposed to U.S. import tariffs.

The data for this analysis was gathered from the Bureau of Economic Analysis and from firms who traded products subject to the tariffs. 5,200 products were classified at the HS-6 level. To estimate the effects of the import tariffs, Handley et al. employed an event-

study framework and a difference in differences estimation. To do this, Handley et al. first created measures of a product's exposure to import tariffs. These measures were broken down into direct and indirect exposure. Direct exposure was measured by looking at firms that are exposed to import tariffs and both import and export goods. This allowed for a connection to be made between these exposed firms and their exports. The other measure of direct exposure looked at firms' exposure to tariffs relative to their variable costs. Next, to estimate indirect exposure, a measure of the exposure of firms that import but do not export was constructed along with another measure of the effect of tariffs on exports through Input-Output (IO) linkages. The first measure of indirect exposure relates to exports since exporters may source their inputs from the tariff exposed importers. The second measure of indirect exposure relates to exports since exporters may purchase products that were subject to tariffs. A difference in differences estimation was then used to learn if exports of products that were exposed to import tariffs were lower after tariffs were levied relative to before. Log exports was regressed on the exposure measures, along with controls for foreign retaliatory tariffs, seasonality, time, sector, and country fixed effects. Finally, an event study that allows estimations to vary by quarter was used to check for pre-trends.

Analysis revealed that there was a drop in exports even after controlling for seasonality and foreign retaliation, and results from the event study revealed no pre-existing trends. When examining products with direct exposure, the difference in differences analysis revealed that products with high exposure to import tariffs were exported less. Also, it was found that for goods with exposure to tariffs on aluminum and steel, the decline in exports was not significant, though exports of products exposed to the rest of the tariffs do decline significantly. The authors theorized this was due to the fact that domestic and non-tariff affected countries were able to meet demand. Looking at goods with high versus low exposure found that goods with high exposure drove the negative effect on exports. The measure using variable costs strengthened the findings of decreased exports. The first measure of indirect

exposure revealed a decline in exports for higher exposure, while the second measure was insignificantly different from zero and therefore does not have much explanatory power. The results of the regression analysis led the authors to conclude that on average, exports were 2.9 log points lower in 2019 relative to the pre-tariff period. Furthermore, the loss in exports was equal to a tariff of 2 percent of the value of total exports. This ad valorem tariff reached 4 percent for the most exposed products. Finally, Handley et al. suggested that firms affected may have had trouble adjusting their supply chains in response to the tariffs.

The impact of foreign retaliatory tariffs on U.S. exports was quantified above by [Amiti et al. \(2019\)](#), [Fajgelbaum \(2020a\)](#), and [Fajgelbaum \(2020b\)](#). What happened as the trade war went on? Was the decline in U.S. exports reallocated to other countries? [Benguria and Saffie \(2025\)](#) provided further estimates of the effects on U.S. exports and their reallocation. The authors also looked at whether Chinese state owned enterprises (SOEs) played a role in shaping the trade war, as they may import less or no U.S. products at the direction of the Chinese government.

Data on trade flows, tariffs, imports by SOEs, Chinese import flows, and others were collected from a wide range of sources. They include [Fajgelbaum et al \(2020a\)](#), confidential firm data from Chinese customs, and the U.S. Census Bureau. U.S. export values, quantities, and unit values were gathered at the HS10 level from January 2016 to December 2019. Industry level measures of U.S. import shares by SOEs were created at the HS-4 level, and were defined as the SOE imports from the U.S. divided by the total imports from the U.S. To quantify the effect of Chinese retaliatory tariffs on U.S. exports, the authors regressed the 12 month change in export values, quantities, and unit values on the 12 month change of one plus the ad-valorem Chinese tariff rates. Product, sector, and time fixed effects were included as-well. A difference in differences regression similar to the one described above was used to estimate the effects of Chinese SOEs on U.S. exports. Other difference in differences regressions, including triple difference specifications, were used to estimate the heterogeneity

of the effect across products, timing of the effects, and several robustness checks. To see if U.S. exporters sent their exports elsewhere, the log of U.S. exports was regressed on 3 terms that capture the effect of U.S. tariffs on ROW destinations, effects of Chinese tariffs, and Chinese SOEs. To determine political effects, a "Republican associated industry" measure was constructed that is the share of Republican votes in 2016 associated with an industry based on the location of its employment.

A 10 percent higher Chinese retaliatory tariff resulted in a 5.6 percent decline in exported value, a 7.1 decline in the quantity of exports, and a 0 percent difference in prices of exports which implied a near complete pass-through to Chinese import prices. U.S. products with a high SOE share fell after accounting for the effect of Chinese retaliatory tariffs. To assess the magnitude of this decline, the authors found that at the average SOE share, U.S. exports fell by 4 percent. The effect on prices was not statistically different from zero, providing additional evidence of pass-through. These results held in several robustness checks; the results were not driven by the oil or soybean industries and related party trade, among others. The authors found that the effect of SOEs was heterogeneous across U.S. sectors. Sectors most affected were food and agriculture along with industrial supplies; there was a 19.4 percent decline in exports of food and agricultural products. It was also determined that decline in exports caused by SOEs may have politically driven. There was a 5 percent decline in exports among "Republican industries" and the coefficient was almost 6 times larger in industries with a high Republican vote share. SOEs had their largest effect during the fourth quarter of 2018, implying that China preferred to use SOEs as a tool after they imposed tariffs on most U.S. goods, and before raising the tariff rate on those goods higher. Regression analysis revealed that U.S. exports affected by retaliatory tariffs were near fully rerouted to other countries, but U.S. exports affected by SOEs were not. The authors theorized that this may be due to the short duration of the SOEs impact and its notability to firms, or that rerouting exports was less feasible during the later stages of the trade war.

VII Conclusion

The 2018 Trade War had vast negative consequences on the U.S. economy. The literature is unanimous in finding near complete pass-through to U.S. import prices, implying that the incidence of the trade war fell on the U.S. This result was notable considering that traditional trade theory would suggest that U.S. tariffs would drive down foreign export prices ([Amiti et al. 2020b](#)). There was also pass-through to U.S. consumers, but that may have not been consistent across goods. This suggests that in the short term, U.S. firms who engaged in front running and trade diversion absorbed the cost of the tariffs in their profit margins. Some goods, like washers and dryers, saw rapid and double digit price increases, along with price hikes by unaffected domestic firms who used their market power to increase prices. There was incomplete pass-through to U.S. producer prices for similar reasons as consumer goods, though price increases of nearly 3 percent were found. Reductions to aggregate real income were between \$7-8 billion in 2018 and grew to about \$24 billion through 2019, equal to a reduction in real GDP of 0.13 percent. The trade war decreased aggregate consumption by approximately \$20 to \$54 billion, with a near 4 percent decline in consumption growth in the most exposed counties. Foreign retaliatory tariffs were consistently found to decrease employment, while import tariffs were found to have small or weak positive effects. The jobs that were gained, as from the washing machine tariffs, cost consumers \$817,000 per job. At best, [Autor et al. \(2024\)](#) found that the effect of the trade war on employment was a wash, if not slightly negative. The literature found extensive evidence of trade diversion, however imports from non-tariff affected countries were not able to replace those lost from China, resulting in a loss of imports overall. Furthermore, U.S. import tariffs caused exports to decline, equal to an ad valorem tariff of 2 percent. While the exports lost from Chinese retaliatory tariffs were rerouted elsewhere, the losses from Chinese SOEs were not, causing a 4 percent decline at the average SOE share. Finally, domestic investment saw around a 1

to 2 percent decline due to losses in capital. Losses in the stock market were found to take \$1.7 trillion off the value of firms sampled.

Further research is needed to fully understand the effects of the trade war. One example can be seen in [Cavallo et al. \(2021\)](#), whose findings on the prices paid by consumers in the short run very well may have changed as time went on. Long term effects are rarely studied in the literature and may uncover new impacts. For example, [Amiti et al. \(2020b\)](#) found that foreign exporters started to lower their prices on steel inputs as time went on. However, as [Autor et al. \(2024\)](#) points out, the COVID pandemic makes the long term effects in their study hard to establish. This is a challenge that faces the literature as a whole. Further research is also needed to definitively determine if U.S. exporters lowered their prices in response to foreign retaliatory tariffs, as the literature conflicts on this point.

As of April 12, 2025, a 10 percent tariff on all global goods, a 145 percent tariff on Chinese goods, and dozens of additional tariffs on numerous countries is in effect ([Bown 2025](#)). What might happen, based on what we observed in the literature, if these and foreign retaliatory tariffs are kept in place? As we saw with [Fajgelbaum \(2020b\)](#), aggregate real income does not fall at a consistent rate with increases in tariffs. The sheer magnitude of the current tariff rates and the value of imports subject to them may result in reductions in welfare several magnitudes larger than before. If any jobs are brought back to the U.S., it will come at a substantial cost. Further, since firms will no longer be able to divert trade to non-tariff affected countries, we may see a more consistent, larger increase in the price of consumer goods as pre-tariff inventory is dwindled down. The current strife in international trade has been much more volatile than before, so it is possible that there may be larger decreases in investment. Overall, the negative impacts seen in 2018 may become much larger. These are only a few possible effects, and the long term impacts of modern trade wars may finally be able to be studied. Indeed, a vast literature will be needed to fully understand the effects of the 2025 Trade War.

References

- Amiti, Mary, Sang Hoon Kong et al. "The Effect of the U.S.-China Trade War on U.S. Investment." *NBER Working Paper Series*, May 2020a. W27114, <https://doi.org/10.3386/w27114>. Working paper.
- Amiti, Mary, Stephen J. Redding, et al. "The Impact of The 2018 Tariffs on Prices and Welfare." *Journal of Economic Perspectives*, vol. 33, no. 4, 1 Nov. 2019, pp. 187–210, <https://doi.org/10.1257/jep.33.4.187>.
- Amiti, Mary, Stephen J. Redding, et al. "Who's paying for the US tariffs? A Longer-Term Perspective." *AEA Papers and Proceedings*, vol. 110, 1 May 2020b, pp. 541–546, <https://doi.org/10.1257/pandp.20201018>.
- Autor, David, et al. "Help for the Heartland? The Employment and Electoral Effects of the Trump Tariffs in the United States." *NBER Working Paper Series*, Jan. 2024. W32082, <https://doi.org/10.3386/w32082>. Working paper.
- Benguria, Felipe, and Saffie, Felipe. "Beyond Tariffs: How Did China's State-Owned Enterprises Shape the US-China Trade War?" *NBER Working Paper Series*, vol. 33599, Mar. 2025, <https://doi.org/10.3386/w33599>. Working Paper.
- Bown, Chad P. "Trump's trade war timeline 2.0: An up-to-date guide." Peterson Institute for International Economics, 11 April 2025, <https://www.piie.com/blogs/realtime-economics/2025/trumps-trade-war-timeline-20-date-guide>
- Bown, Chad P, and Melina Kolb. "Trump's Trade War Timeline: An up-to-Date Guide." Peterson Institute for International Economics, 20 Jan. 2025, www.piie.com/sites/default/files/documents/trump-trade-war-timeline.pdf.
- Caldara, Dario, et al. "The Economic Effects of Trade Policy Uncertainty." *Journal of Monetary Economics*, vol. 109, Jan. 2020, pp. 38–59, <https://doi.org/10.1016/j.jmoneco.2019.11.002>.
- Cavallo, Alberto, et al. "Tariff Pass-Through at the Border and at the Store: Evidence from US trade policy." *American Economic Review: Insights*, vol. 3, no. 1, 1 Mar. 2021, pp. 19–34, <https://doi.org/10.1257/aeri.20190536>.
- Dang, Alicia, et al. "Winners and losers from the U.S.-China Trade War." *NBER Working Paper Series*, Nov. 2023. W31922, <https://doi.org/10.3386/w31922>. Working paper.
- Fajgelbaum, Pablo D, et al. "The Return to Protectionism." *The Quarterly Journal of Economics*, vol. 135, no. 1, Feb. 2020a, pp. 1–55, <https://doi.org/10.1093/qje/qjz036>.
- Fajgelbaum, Pablo D., et al. "Updates to Fajgelbaum et al. (2020) with 2019 Tariff Waves". Princeton University, 21 Jan. 2020b. <http://www.econ.ucla.edu/pfajgelbaum/rtpupdate.pdf>.

- Flaaen, Aaron, et al. “The Production Relocation and Price Effects of US Trade Policy: The Case of Washing Machines.” *American Economic Review*, vol. 110, no. 7, July 2020, pp. 2103–2127, <https://doi.org/10.1257/aer.20190611>.
- Flaaen, Aaron, and Justin Pierce. “Disentangling the Effects of the 2018-2019 Tariffs on a Globally Connected U.S. Manufacturing Sector.” *Finance and Economics Discussion Series*, 2019.0, no. 86, Dec. 2019, <https://doi.org/10.17016/feds.2019.086>.
- Handley, Kyle, et al. “Rising Import Tariffs, Falling Exports: When Modern Supply Chains Meet Old-Style Protectionism.” *American Economic Journal: Applied Economics*, vol. 17, no. 1, Jan. 2025, pp. 208–238, <https://doi.org/10.1257/app.20210051>.
- Nicita, Alessandro. “Trade and Trade Diversion Effects of United States Tariffs on China.” UNCTAD Research Paper, vol. 27, Nov. 2019, <https://unctad.org/publication/trade-and-trade-diversion-effects-united-states-tariffs-china>.
- Waugh, Michael. “The Consumption Response to Trade shocks: Evidence From The US-China Trade War.” *NBER Working Paper Series*, vol. w26353, Dec. 2019, <https://doi.org/10.3386/w26353>. Working Paper.